Umer Farooq Ahmed, PhD

Assistant Professor, Capital University of Science and Technology

Nanoelectronics
FET Modeling and Simulation
Large Area Electronics
Sputtering
Nanowire Devices

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ZnO Nanowire Gate-Tunable Schottky Diodes by Adhesion Lithography

Umer F. Ahmed and Andrew J. Flewitt

***Abstract*— Electronic devices using nanomaterials have the potential to outperform equivalent devices using bulk materials, but only if nanoscale contacts can be made to the nanomaterial. This is achieved by a scalable process in this work through hydrothermal growth of horizontal ZnO nanowires (NWs) from a ZnO seed layer across a 40 nm nanogap created by adhesion lithography (A-lith). The asymmetric Ti and Au electrodes of the nanogap lead to the formation of Schottky diodes in the NWs, exhibiting rectification ratios $>10^3$. The addition of SiO_2 dielectric around the NW improved the rectification ratio to almost 10^4 and allowed the addition of a bottom gate. The device demonstrated transistor-like behavior, most notably under reverse bias conditions. These characteristics are analogous to a source-gated transistor (SGT), with both accumulation and depletion modes of operation, demonstrating for the first time experimentally, SGTs with a size below 100 nm. This is significant as it opens up the possibility of nanoscale arrays of SGTs for ultra-high resolution displays or neuromorphic processors.**

***Index Terms*— Zinc oxide, Nanowires, Nano-electronics, Adhesion lithography, Source gated transistors, Neuromorphic computing**

I. Introduction

There is significant interest in producing nanoscale arrays of transistors on flexible and unconventional substrates for high resolution virtual reality headset displays and neuromorphic processors [1], [2]. Integration of nanomaterials into electronic devices require the creation of electrical contacts at the nanoscale to utilize their intrinsic properties. This usually involves electron beam microscopy/lithography to visualize the material and pattern the contacts, which is not a scalable process [3]. In this work, scalable processing is achieved through the combination of two bottom-up fabrication processes: hydrothermal growth of ZnO NWs [4] is combined with A-lith for coplanar electrode patterning [5].

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Umer F. Ahmed and Andrew J. Flewitt are with the Electrical Engineering Division, Department of Engineering, Cambridge University, CB3 0FA Cambridge, U.K. (Corresponding author:.)

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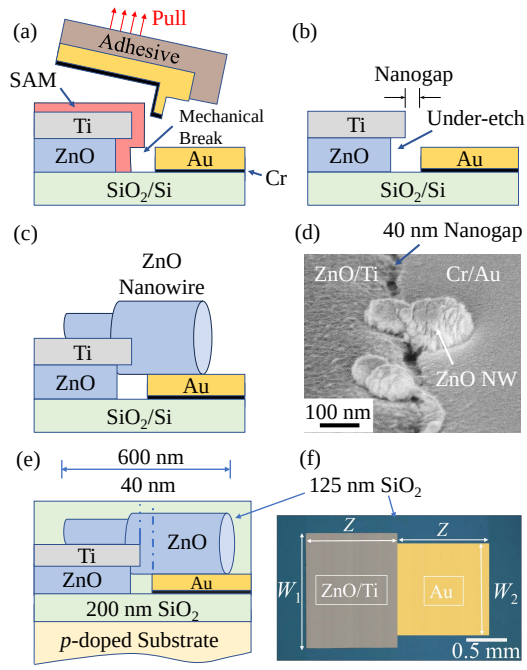

Fig. 1. Schematic diagram of: a)-b) nanogap fabrication using A-lith and c) NW growth inside the nanogap. d) SEM image of ZnO NW Schottky device after 10 min in NW growth solution. e) Schematic diagram and f) optical micrograph of a bottom gated ZnO NW Schottky device.

A-lith uses self-assembled monolayers (SAMs) to alter the surface adhesion between electrodes, which can be peeled from one another to create coplanar nanogaps ranging from a few nanometers to tens of nanometers [5]. The technique does not suffer from limitations such as poor scalability or electrode and substrate materials, commonly seen in direct-write lithography techniques. It opens up a future avenue for nanoelectronics, wherein integrated nanodevices can be fabricated without specialized equipment.

A range of nanodevices for large-area electronic (LAE) applications fabricated using A-lith have been reported previously, including Schottky diodes, thin-film transistors, and memristors [6]–[11]. Although A-lith has been mostly used to make thin film or solution-processed devices, the technique has potential to be employed in bottom-up nanomaterial device fabrication, as illustrated in [12]. This letter combines A-lith with bottom-up NW growth to make gate-tunable self-aligned NW Schottky devices suitable for LAEs by incorporating a seed layer for NW growth in the device geometry. The result is the first experimental demonstration of nanoscale SGT.

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Article

"Nano-In-Nano" Schottky Diodes Fabricated by Combining Self-Aligned Nanogap Patterning with Bottom-Up ZnO Nanowire Growth

Umer Farooq Ahmed, Gwenhivir S. Wyatt-Moon, and Andrew J. Flewitt*

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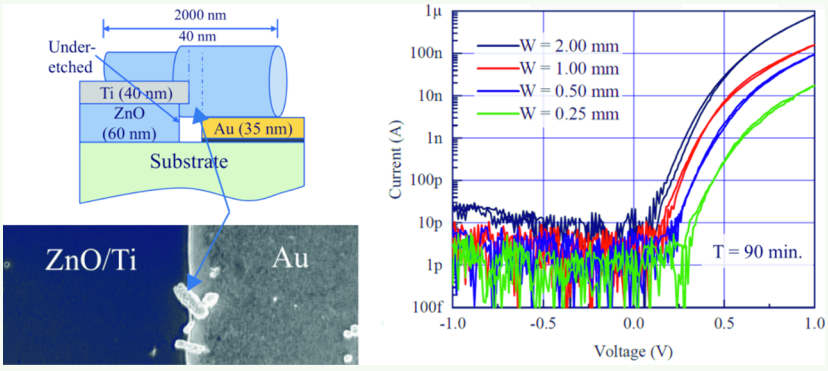

ABSTRACT: Nanoscale semiconductors offer significant advantages over their bulk semiconductor equivalents for electronic devices as a result of the ability to geometrically tune electronic properties, the absence of internal grain boundaries, and the very low absolute number of defects that are present in such small volumes of material. However, these advantages can only be realized if reliable contacts can be made to the nanoscale semiconductor using a scalable, low-cost process. Although there are many low-cost "bottom-up" techniques for directly growing nanomaterials, the fabrication of contacts at the nanoscale usually requires expensive and slow techniques like e-beam lithography that are also hard to scale to a level of throughput that is required for commercialization. A scalable method of fabricating such devices is demonstrated in this work by combining two bottom-up fabrication techniques. ZnO nanowire Schottky diodes are produced with a device length of a few tens of nanometers and a performance significantly exceeding a ZnO thin film equivalent. The first technique is adhesion lithography that allows self-aligned coplanar electrodes of different materials to be patterned with a nanogap ∼10 to 50 nm length between the two. In this case, one electrode is gold, while the other is a bilayer of titanium on a thin film of ZnO, and it is this thin film that allows the second technique, hydrothermal growth, to be used to grow ZnO nanowires directly across the nanogap. The resulting "nano-in-nano" Schottky diodes have a high rectification ratio $>10^4$, a low turn-on voltage <0.3 V, and a minimal off-state current <10 pA. This process could be used to realize a variety of nano-in-nano electronic devices in the future, including short channel gate-all-around (GAA) transistors.

KEYWORDS: *ZnO nanowires, Schottky diodes, hydrothermal growth, adhesion lithography, self-assembled monolayer, large-area electronics, nanoelectronics, bottom-up fabrication*

■ INTRODUCTION

The increase in the computational capability of microprocessors and the ability to store vast quantities of information in small memory devices has enabled the mobile device revolution of the last two decades; this has been achieved by a relentless move to ever smaller fabrication nodes for silicon CMOS fabrication. However, a point is being reached where simple scaling down of transistor channel length to increase system performance is no longer effective. In particular, new device geometries and stacking of devices in three dimensions will be required in future.[1] In some ways, the introduction of the FINFET structure in mainstream CMOS processing in the early 2000s is an example of this; the traditional "planar" semiconducting channel was replaced with vertically oriented semiconducting fins with a

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An improved model to predict DC characteristics of organic field-effect transistors

Muhammad Mansoor Ahmed[1] · Khasan Karimov[2] · Umer Farooq Ahmed[1]

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Abstract
Organic field-effect transistors (OFETs) are becoming popular because of the ease associated with their fabrication and cost-effectiveness. Numerous models have been developed to predict the room temperature $I - V$ characteristics of OFETs involving their design, as well as material parameters. In this paper, we discuss efforts that have been made to develop a compact model which can be employed by design engineers with reasonable ease without compromising the accuracy of modeled characteristics. Both for linear and for saturation region of $I - V$ characteristics, a single expression involving a hyperbolic tangent function has been proposed. The proposed model addresses typically observed contact resistances of drain–source electrodes by incorporating the effective drain-to-source potential and thus generates OFETs output characteristics with improved accuracy. A modification in the device threshold voltage is proposed to improve modeled data of short channel OFETs. It has been demonstrated by evaluating root mean squared error (RMSE) values that the proposed technique is at least 15% better than the best reported model of OFETs.

Keywords FinFET · Heterojunction devices · DC characteristics · Analytical model

1 Introduction

Organic semiconductors (OSCs) have numerous applications for electronic and optoelectronic devices, such as light-emitting diodes (LEDs), field-effect transistors (FETs), and photovoltaics (PVs). Most importantly, a considerable effort has been made in the development of organic materilas to improve FETs performance. Furthermore, OSC-based FETs are considered to be one of the key components of organic integrated circuits [1, 2]. Organic field-effect transistors (OFETs) have attracted a significant attention in microelectronics industry due to their low fabrication cost, compatibility with large area printing technology, good solution processability, and low temperature deposition.

In order to establish OFET integration with respective circuitry, there is a need to have an OFET model, which can predict the device characteristics under varying operating conditions. For efficient handling, the model should be based on the device physical parameters and the same should not be a complicated one. Since there is a variety of organic materials, which can be employed to fabricate OFETs, the model should be technically sound to simulate the device characteristics independent of OSC. It should be compact to be employed in CAD-related tools meant for OFET-based circuit design. Additionally, the model should be computationally efficient with the ability to predict the device characteristics with high accuracy.

Meixner et al., in 2008 [3], reported an OFET model, which was primarily based on MOSFET expressions. These expressions were modified under the assumption that OFET channel conductivity would depend upon the organic film used in its fabrication. They suggested a modification in MOSFET expression with a term, which they referred to as voltage-dependent bulk conductance. Furthermore, they suggested a field-dependent mobility term to allow the model to track the experimental data with better accuracy. In general,

✉ Muhammad Mansoor Ahmed
mansoor@cust.edu.pk

Khasan Karimov
khasansangink@gmail.com

Umer Farooq Ahmed
umerfarooq_62@hotmail.com

1 Capital University of Science and Technology, Islamabad, Pakistan

2 GIK Institute of Engineering Sciences and Technology, Topi, Sawbi, Pakistan

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A Unified Depletion/Inversion Model for Heterojunction Trigate FinFETs DC Characteristics

QAMAR-UD-DIN MEMON, UMER F. AHMED, (Member, IEEE), AND MUHAMMAD M. AHMED, (Senior Member, IEEE)
Department of Electrical Engineering, Capital University of Science and Technology, Islamabad 44000, Pakistan
Corresponding author: Muhammad M. Ahmed (mansoor@cust.edu.pk)

ABSTRACT This paper presents a comprehensive $I-V$ model for the evaluation of above threshold DC characteristics of trigate AlGaN/GaN FinFETs for both depletion and inversion mode of operations. For the depletion mode, 2DEG carrier concentration, n_s, is evaluated considering the trigate geometry of the device, which is then used to assess drain current (I_{2D}) for both the linear as well as the saturation region of operations. At a relatively higher gate bias, where the 2DEG is fully un-depleted, a significant rise in the drain current is observed which is associated with the creation of two additional side channels in the GaN layer of AlGaN/GaN FinFET, caused by the inversion of carriers. This component of the drain current, referred to as inversion current (I_{inv}), is modeled using the solution of a 2D Poisson equation. The total current (I_{ds}) is the summation of both the depletion as well as the inversion currents ($I_{ds} = I_{inv} + I_{2D}$). The developed technique is validated using experimental data and is seen to exhibit a good degree of accuracy. Therefore, the proposed model could serve as a useful tool to predict AlGaN/GaN FinFETs output and transfer characteristics.

INDEX TERMS FinFET, heterojunction devices, DC characteristics, analytical model.

NOMENCLATURE

x, y, z	Co-ordinate system
2D, 3D	2 and 3 dimensional
2DEG	2 dimensional electron gas
AlGaN	Aluminum gallium nitride
GaN	Gallium nitride
C_{ox}	Oxide capacitance
d	Distance between top gate and 2DEG
E_F	Fermi level energy
E	Field inside the channel
E_{sat}	Saturation field
ϵ_{eff}	Permittivity of 2DEG
ϵ_A	AlGaN permittivity
ϵ_G	GaN permittivity
H_{fin}	Fin height
I_{ds}	Drain to source current
I_{inv}	Inversion current
I_{2D}	Current from two dimensional electron gas
k_B	Boltzmann constant
L_g	Gate length
$L_{g,W}$	Gate length and width
N_d	Bulk concentration
n_s	2DEG carrier concentration
n_i	Intrinsic carrier concentration
q	Electronic charge
Q_i	Inversion charges
T	Temperature
T_{ox}	Oxide thickness
T_{fin}	Fin thickness
V_s	Source voltage
V_d	Drain voltage
V_g	Gate voltage
V_{gs}	$V_g - V_s$
V_{ds}	$V_d - V_s$
V_T	Thermal voltage
V_{fb}	Flat band voltage
V_{th}	Threshold voltage
$V_{d(sat)}$	Drain voltage at saturation point
V	Applied voltage
v_{sat}, v_d	Saturation and drift velocity
ϕ_b	Built in potential
ψ	Surface potential
ψ_0	Mid channel potential
μ_0	Low field mobility
μ	Mobility
ΔE_c	Bandgap discontinuity
ΔL_g	Channel length modulation

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A modified analytical model for AlGaN/GaN FinFETs $I-V$ characteristics

U F Ahmed, M M Ahmed and Q D Memon

Department of Electrical Engineering, Capital University of Science and Technology, Islamabad, Pakistan

E-mail: mansoor@cust.edu.pk

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Abstract

In this paper, a high electron mobility transistor's (HEMT) analytical model, which is based on the vertical operation of Schottky barrier gate, has been modified for the $I-V$ characteristics of rectangular shaped AlGaN/GaN FinFETs having three-sided Schottky barrier gate operation. The proposed model includes the effect of the tri-gate structure on sheet carrier concentration (n_s) of the device. A three dimensional Poisson equation is solved keeping in view, the device geometry and applied potentials, to get the effect of the side gates on n_s. It is demonstrated that n_s of a FinFET depletes relatively faster than its HEMT counterpart due to the extra fields caused by the side gates. Knowing bias dependent n_s, FinFET $I-V$ expressions are developed and tested on AlGaN/GaN FinFETs of varying gate lengths ($L_g = 0.4 - 1.0\ \mu$m) for the above threshold regime. A good agreement between the experimental and modeled characteristics is observed, which demonstrates the validity of the proposed model in predicting the DC characteristics of tri-gate AlGaN/GaN FinFETs.

Keywords: FinFET, analytical model, DC characteristics, heterostructure

(Some figures may appear in colour only in the online journal)

Nomenclature

x, y, z	Rectangular coordinates
n_s	2DEG carrier concentration
ϵ_s	Permittivity of semiconductor
d	Top gate to 2DEG distance
V_s	Source voltage
V_d	Drain voltage
V_g	Gate voltage
V_{ds}	$V_d - V_s$
V_{gs}	$V_g - V_s$
V_{th}	Threshold voltage
ψ	Spatial potential distribution
ψ_{ds}	Surface potential due to V_{ds}
ψ_{gs}	Surface potential due to V_{gs}
ΔE_c	Bandgap discontinuity at heterojunction interface
E_F	Fermi energy
E_y	y-directed electric field
E_{sat}	Saturation electric field
ϕ_b	Schottky barrier height
N	Doping density
q	Electronic charge
σ	Polarization controlled charge density
μ_0	Low field mobility
v_d	Carrier velocity
v_{sat}	Carrier saturation velocity
γ_i	Adjustment factors
I_{ds}	Drain to source current
T_{fin}	Thickness of fin
L_g	Gate length
H_{fin}	Height of fin
ΔL_g	Gate length modulation
W	Width of fin

1. Introduction

The downscaling of field effect transistor (FET) is a perpetual pursuit of device shrinkage to meet the growing requirements

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